

Documentation of Changes Resulting from Testing

HealthConnect Platform Validation & Operational Enhancements

Project: HealthConnect Testing & Validation Framework | **Scope:** Dataset Audit & System Optimization

Following rigorous programmatic validation, data integrity audits, and segment-level consistency checks of the HealthConnect appointment dataset, several key modifications and operational optimizations were implemented across the framework:

1. Refined Visual Scalability & Layout

Action & Impact

Removed cluttered multi-series charts in favor of clean, segmented bar plots and standardized metric cards. This eliminated visual ambiguity and improved leadership "at-a-glance" comprehension of core attendance KPIs.

2. Enforced Data Hygiene & Null-Handling Rules

Action & Impact

Implemented strict list-wise exclusion protocols and explicit null-handling for secondary attributes (reminder_channel, distance_to_clinic_km, waiting_time_minutes), preventing metric distortion caused by unrecorded fields.

3. Transition to Dynamic Intervention Thresholds

Action & Impact

Replaced static reminder schedules with a milestone-based multi-touch cadence (14-day, 7-day, and 48-hour check-ins) informed by confirmed booking lead-time concentration curves to combat patient disengagement.

4. Integration of Geospatial & Accessibility Adjustments

Action & Impact

Added distance-to-clinic risk factors into scheduling parameters to identify high-travel-barrier segments, preparing the operational framework to route telehealth alternatives for patients residing beyond standard proximity thresholds (>10 km).